

Nikola Tesla

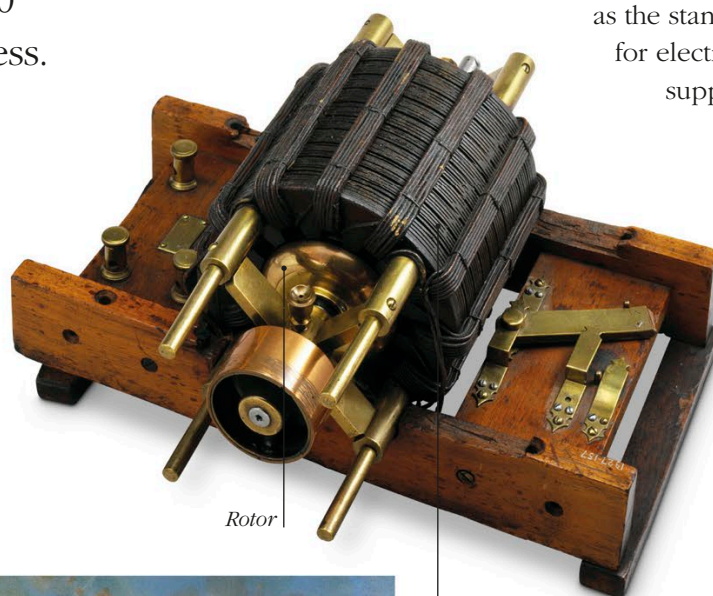
The gifted inventor Nikola Tesla produced the first motor that ran efficiently on an alternating current, experimented with X-rays, and demonstrated a radio-controlled boat. During his life, he held about 300 patents, but ended up virtually penniless.

UNLUCKY GENIUS

Tesla claimed he had a photographic memory and that many of his ideas came to him fully formed. However, he was not a canny businessman. After moving to New York City, Thomas Edison hired the young Tesla, and offered to pay him \$50,000 for an improved design of a motor. When Tesla presented a solution and asked for the money, Edison replied that he was only joking.

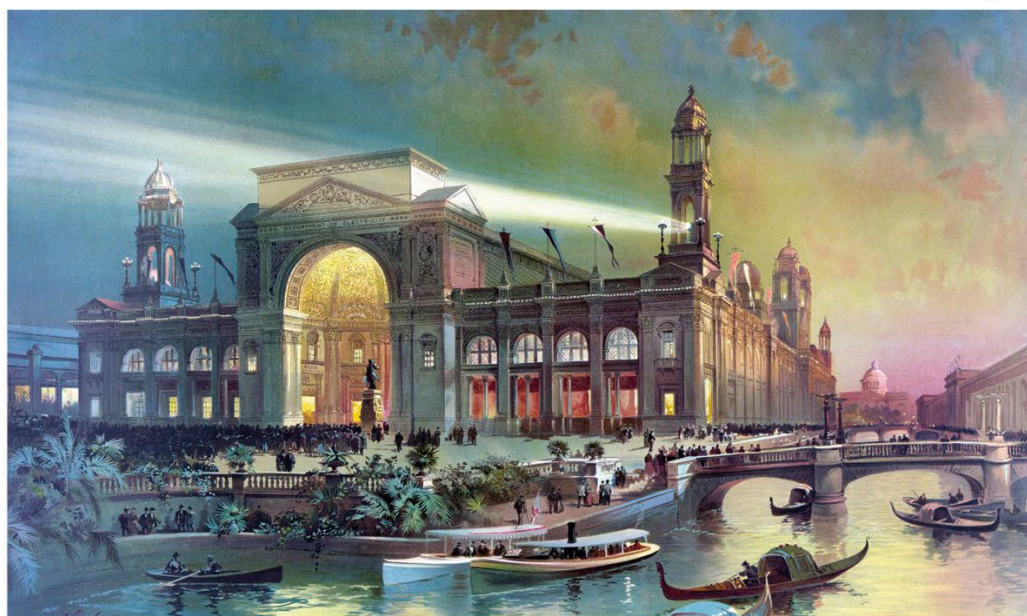
INDUCTION MOTOR

In 1887, Tesla developed an induction motor (below) that ran on alternating current (AC)—a power system that performed better than the existing direct current (DC) in long-distance, high-voltage transmission. This device led to the adoption of AC as the standard for electricity supplies.



Rotor

Stator generates rotating magnetic field that turns the rotor.



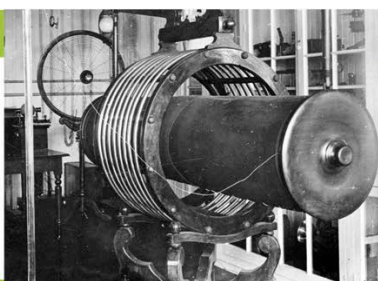
LIGHTING UP

Tesla and his financial backer Westinghouse were in a “war of currents” with Thomas Edison to promote their rival systems. In 1893, Tesla won the bid to light that year’s World’s Columbian Exhibition (left). His success was key to proving the reliability of AC electrical power.

1891

He invents the Tesla coil, which would be widely used in radio technology.

Tesla coil, 1895



1898

At New York’s Madison Square Garden, Tesla gives a public demonstration of a boat guided by remote control that uses radio signals.

1943

Tesla dies in New York City, aged 86. The AC system that he developed remains the global standard for power transmission.